

Following examples showing graphical methods to find different locations of Quartiles, Deciles and Percentiles with appropriate data values.

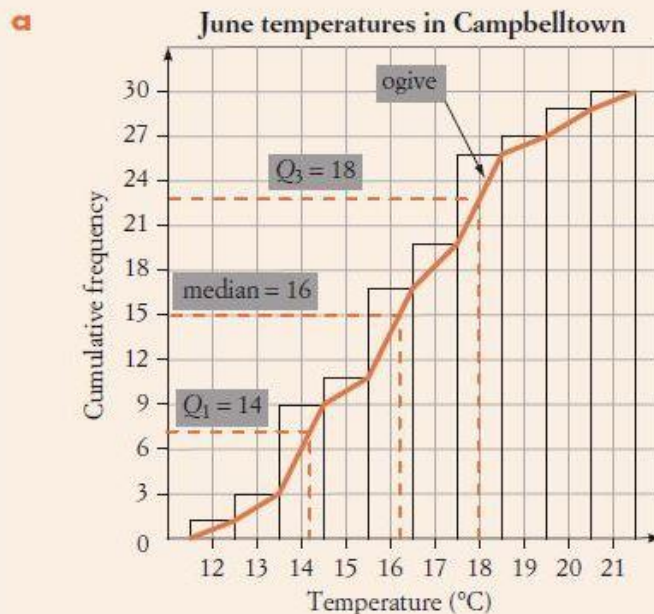
EXAMPLE 14

The maximum daily temperatures (in °C) in Campbelltown in June were recorded and grouped into the frequency table shown.

- Draw a cumulative frequency histogram and polygon for the data.
- Use the frequency polygon to find the median and calculate the interquartile range.

Temperature (°C)	Frequency	Cumulative frequency
12	1	1
13	2	3
14	6	9
15	2	11
16	6	17
17	3	20
18	6	26
19	1	27
20	2	29
21	1	30

Solution



The ogive (polygon) is always inside the columns.

- b** Draw a horizontal line from the halfway mark (15) on the cumulative frequency axis to where it meets the ogive. The median is the corresponding value on the 'Temperature' axis.

Median = 16

To find Q_1 , draw a horizontal line from the quarter mark ($\frac{1}{4} \times 30 = 7.5$) on the cumulative frequency axis to where it meets the ogive, then read the temperature value.

$Q_1 = 14$

To find Q_3 , draw a horizontal line from the three-quarter mark ($\frac{3}{4} \times 30 = 22.5$) on the cumulative frequency axis.

$Q_3 = 18$

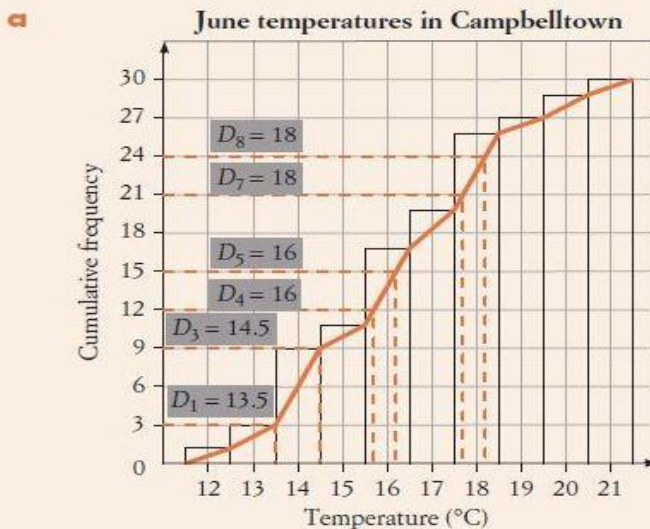
$$\begin{aligned}\text{Interquartile range} &= Q_3 - Q_1 \\ &= 18 - 14 \\ &= 4\end{aligned}$$

EXAMPLE 15

Use the cumulative frequency graph from Example 14 to answer the following questions.

- a** Find:
- the 4th decile, D_4
 - the 7th decile, D_7 .
- b** What value cuts off the top 20% of temperatures?
- c** Between which two deciles would you find a temperature of 14°C ?

Solution



The deciles are marked at intervals of three units on the cumulative frequency axis.

i $D_4 = 16$

ii $D_7 = 18$

b D_8 cuts off the top 20% of temperatures, so the value is 18.

c Between D_1 and D_3 .

EXAMPLE 16

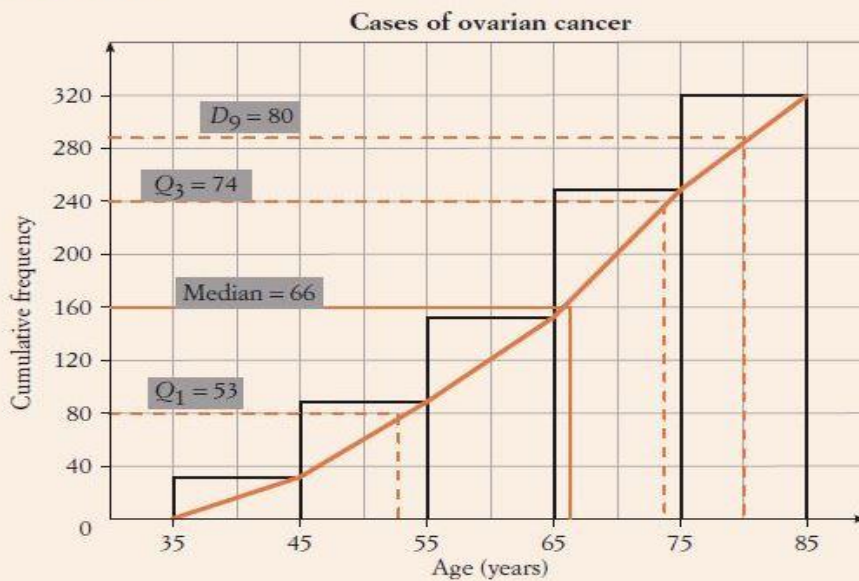
The number of cases of ovarian cancer in women from various age groups is shown below.

Age (years)	Class centre	Frequency	Cumulative frequency
35–<45	40	28	28
45–<55	50	61	89
55–<65	60	65	154
65–<75	70	92	246
75–<85	80	74	320

Draw an ogive for this data and use it to find an estimate for:

- a the median
- b the 3rd quartile
- c the 9th decile
- d the interquartile range.

Solution



All these values are estimates because the

- a** Halfway point on the 'Cumulative frequency' axis = 160
Median ≈ 66

Estimating from the 'Age' axis.

data has been grouped
into class intervals.

- b** The three-quarter point on the 'Cumulative frequency' axis = $\frac{3}{4} \times 320 = 240$
 $Q_3 \approx 74$

- c** 90% point on the 'Cumulative frequency' axis = 0.9×320
 $= 288$

$$D_9 \approx 80$$

- d** Quarter point on the 'Cumulative frequency' axis = $\frac{1}{4} \times 320$
 $= 80$

$$Q_1 \approx 53$$

$$\begin{aligned}\text{Interquartile range} &= Q_3 - Q_1 = 74 - 53 \\ &= 21\end{aligned}$$

